

Collaborators:

Understanding the Superposition Principle

- a. Verify that $y = c_1 y_1 + c_2 y_2$ is also a solution. Discuss why this can be also the general solution.
- b. Given the initial conditions $y(0) = 0$ and $y'(0) = 1$, show that the constants are $c_1 = -1$ and $c_2 = 1$.
- c. Write the specific solution that satisfies the initial conditions in Part (b).
- d. How do initial conditions help us determine the constants in a linear combination of independent solutions?

For a linear 2nd-order non-homogeneous ODE, if y_h is the homogeneous solution and y_p is the particular solution, then by the *superposition principle*, the general solution is of the form

Consider the same ODE from Problem (1) but now the right hand side is now non-zero,

We have shown that the solution to the associated homogeneous equation, that is $y'' + 3y' + 2y = 0$, is

The proposed particular solution is

a. Verify that the general solution is $y = y_h + y_p$.	b. Given the initial conditions $y(0) = 0$ and $y'(0) = 1$, show that the constants are $c_1 = -\frac{1}{2}$ and $c_2 = 0$.
c. Write the specific solution that satisfies the initial conditions in Part (b).	d. Why did the constants change when the equation became non-homogeneous?